## SWITCHBOARD: MDP (EXISTING)

| | |
|---|---|
| **FED FROM:** | POLE MOUNTED UTILITY TRANSFORMER |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 800A |
| **MAIN:** | MCB |
| **AIC:** | 22,000A |

| CKT NO | LOAD DESCRIPTION | LOAD VA | LOAD TYPE | FRAME SIZE | TRIP RATING | # OF POLES | NOTES | PHASE |
|---|---|---|---|---|---|---|---|---|
| 1 | PANEL "C" | 72060 | R | 400A | - - | 3 | | A |
| 2 | PANEL "A" SECTION 1 AND 2 | 112500 | E | 225A | - - | 3 | | B |
| 3 | PANEL "B" | 36030 | R | 200A | - - | 3 | | C |
| **4** | **CHILLER** | **39777** | **M** | **175A** | **- -** | **3** | **1** | **A** |
| **5** | **PANEL "PNL 4"** | **41803** | **0** | **150A** | **- -** | **3** | **1** | **B** |
| 6 | OUTSIDE LIGHTING/POLE LIGHT BY DUMPSTER | 2500 | L | 20A | - - | 2 | | C |
| 7 | SPARE | | | 100A | - - | 3 | | A |
| 8 | POLE LIGHTS | 2500 | L | 40A | - - | 1 | | B |
| 9 | 3-POLE SPACE | | | | | 3 | | C |
| 10 | 3-POLE SPACE | | | | | 3 | | A |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C |
|---|---|---|---|
| LIGHTING | 0 | 2500 | 2500 |
| RECEPTACLES | 72060 | 0 | 36030 |
| MOTORS | 39777 | 0 | 0 |
| LARGEST MOTOR | 0 | 0 | 0 |
| EQUIPMENT | 0 | 112500 | 0 |
| APPLIANCES | 0 | 0 | 0 |

| SUBFEED TOTAL | SWITCHBOARD TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|
| 0 | 5000 | 125% | 6250 |
| 0 | 108090 | 55% | 59045 |
| 0 | 39777 | 100% | 39777 |
| 0 | 0 | 125% | 0 |
| 0 | 112500 | 100% | 112500 |
| 0 | 0 | 100% | 0 |
| | **TOTAL DEMAND LOAD (KVA) =** | | **257.3** |
| | **TOTAL DEMAND LOAD (A) =** | | **714** |

**GENERAL NOTES:**

A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

C. INDICATE SPARES IN PENCIL.

**SPECIFIC NOTES:**

1. LOAD AFFECTED THIS CONTRACT WITH EQUIPMENT REPLACEMENT. LOAD ADDED IS EQUAL OR LESS THAN LOAD REMOVED. NO NEW LOAD ADDED THIS CONTRACT.

## PANEL: PNL 4 (EXISTING)

| | |
|---|---|
| **FED FROM:** | MDP |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 250A |
| **MAIN:** | MLO |
| **AIC:** | 10,000A |

| CKT NO | NOTES | DESCRIPTION | LOAD VA | LOAD TYPE | BREAKER POLE | BREAKER AMPS | PHASE | BREAKER AMPS | BREAKER POLE | LOAD TYPE | LOAD VA | DESCRIPTION | NOTES | CKT NO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | GLYCOL PUMPS | 545 | M | 1 | 20 | A | | | M | 1105 | | | 2 |
| 3 | 2 | **HEAT TRACE** | **1000** | **E** | **1** | **20** | B | 35 | 3 | M | 1105 | HEATING WATER PUMP | | 4 |
| 5 | | BOILER CONTROLS | 545 | E | 1 | 20 | C | | | M | 1105 | | | 6 |
| 7 | | TEMP CONTROL PANEL | 200 | E | 1 | 20 | A | | | M | 1105 | | | 8 |
| 9 | | | **1233** | **M** | | | B | 35 | 3 | M | 1105 | HEATING WATER PUMP | | 10 |
| 11 | **1** | **CHILLED WATER PUMP** | **1233** | **M** | 3 | 35 | C | | | M | 1105 | | | 12 |
| 13 | | | **1233** | **M** | | | A | 20 | 1 | | | SPARE | | 14 |
| 15 | | | 4140 | | | | B | 20 | 1 | | | SPARE | | 16 |
| 17 | | PANEL FCU | 5260 | | 3 | 60 | C | 20 | 1 | M | 1176 | BLOWER FAN | | 18 |
| 19 | | | 5120 | | | | A | 20 | 1 | M | 1176 | MAKE UP AIR UNIT | | 20 |
| 21 | | | 396 | M | | | B | | | M | 396 | | | 22 |
| 23 | | HEATING WATER PUMP | 396 | M | 3 | 15 | C | 15 | 3 | M | 396 | HEATING WATER PUMP | | 24 |
| 25 | | | 396 | M | | | A | | | M | 396 | | | 26 |
| 27 | | WATER HEATER | 200 | E | 1 | 20 | B | 20 | 1 | | | SPARE | | 28 |
| 29 | | EXHAUST FAN | 696 | M | 1 | 15 | C | 20 | 1 | | | SPARE | | 30 |
| 31 | | SPARE | | | 1 | 20 | A | 20 | 1 | E | 100 | FIRE/SMOKE DAMPERS | | 32 |
| 33 | | ELECTRICAL/WORK ROOM | 540 | R | 1 | 20 | B | 20 | 1 | R | 180 | ROOF RECEPTACLE | | 34 |
| 35 | | STORAGE/HALL LIGHTS | 750 | L | 1 | 20 | C | 20 | 1 | | | SPARE | | 36 |
| 37 | | WORK ROOM | 360 | R | 1 | 20 | A | | | **M** | **1585** | | | 38 |
| 39 | | WILKINS OFFICE 309 | 540 | R | 1 | 20 | B | 35 | 3 | **M** | **1585** | **CHILLED WATER PUMP** | **1** | 40 |
| 41 | | R-MANAGEMENT HALL 307 | 720 | R | 1 | 20 | C | | | **M** | **1585** | | | 42 |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C | SUBFEED TOTAL | PANEL TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|---|---|---|---|
| LIGHTING | 0 | 0 | 750 | 0 | 750 | 125% | 937.5 |
| RECEPTACLES | 360 | 1260 | 720 | 0 | 2340 | 100% | 2340 |
| MOTORS | 10448 | 9847 | 11579 | 0 | 31874 | 100% | 31874 |
| LARGEST MOTOR | 1233 | 1233 | 1233 | 0 | 3699 | 125% | 4623.75 |
| EQUIPMENT | 300 | 1200 | 528 | 0 | 2028 | 100% | 2028 |
| APPLIANCES | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| | | | | | **TOTAL DEMAND LOAD (KVA) =** | | **41.8** |
| | | | | | **TOTAL DEMAND LOAD (A) =** | | **116** |

**GENERAL NOTES:**

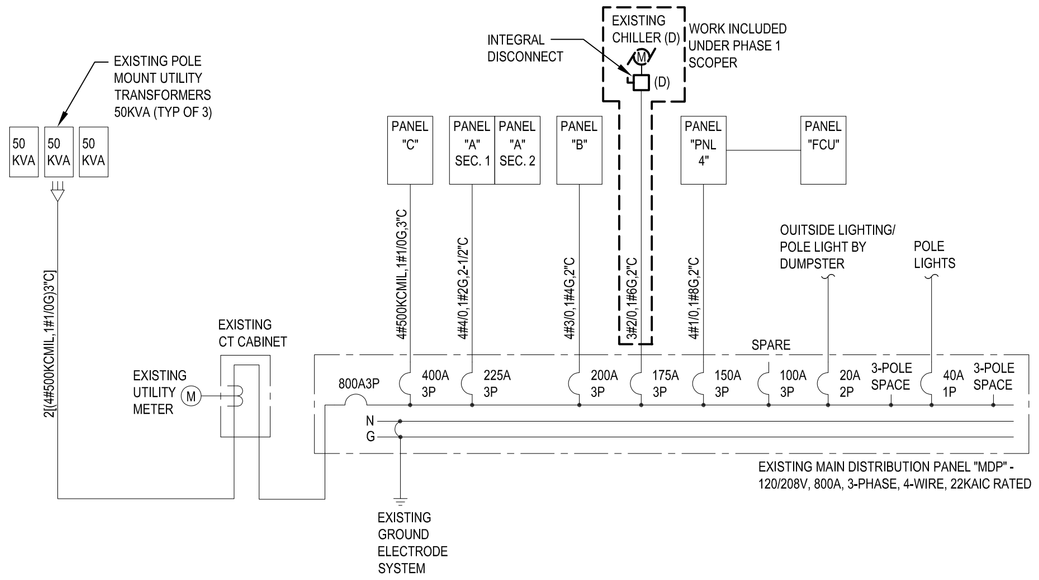
A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

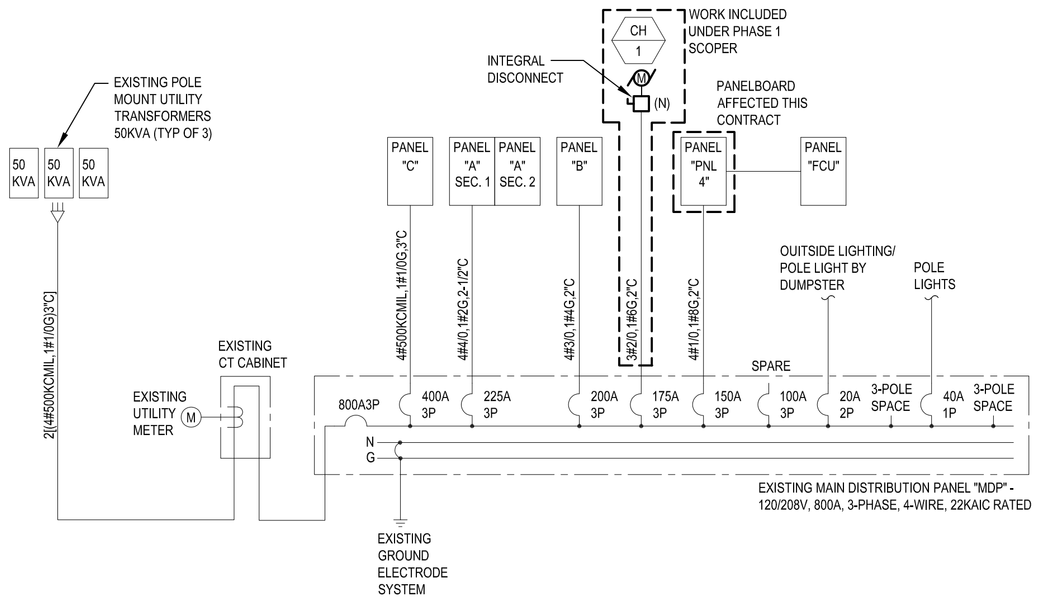
**SPECIFIC NOTES:**

1. CIRCUIT AFFECTED THIS CONTRACT BY EQUIPMENT REPLACEMENT. LOAD ADDED IS EQUAL TO OR LESS THAN LOAD REMOVED.

2.REPLACE EXISTING 20A1P CIRCUIT BREAKER WITH NEW GFEP CIRCUIT BREAKER WITH 30ma TRIP SETTING TO COMPLY WITH 2023 NEC REQUIREMENTS.




1 E500 ELECTRICAL ONE-LINE DIAGRAM (DEMOLITION)
SCALE: NONE




1 E500 ELECTRICAL ONE-LINE DIAGRAM (NEW WORK)
SCALE: NONE

## PANEL: FCU (EXISTING)

| | |
|---|---|
| **FED FROM:** | PNL 4 |
| **LOCATION:** | SEE PLANS |
| **MOUNTING:** | SURFACE |
| **ENCLOSURE:** | NEMA 1 |

| | |
|---|---|
| **VOLTAGE:** | 208Y/120V, 3PH, 4W |
| **BUS AMPACITY:** | 125A |
| **MAIN:** | MLO |
| **AIC:** | 10,000A |

| CKT NO | NOTES | DESCRIPTION | LOAD VA | LOAD TYPE | BREAKER POLE | BREAKER AMPS | PHASE | BREAKER AMPS | BREAKER POLE | LOAD TYPE | LOAD VA | DESCRIPTION | NOTES | CKT NO |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | FAN COIL UNIT 12 | 1080 | M | 1 | 20 | A | 20 | 1 | M | 100 | FAN COIL UNIT 7 | | 2 |
| 3 | | FAN COIL UNIT 13 | 1080 | M | 1 | 20 | B | 20 | 1 | M | 1080 | FAN COOIL UNIT 8 | | 4 |
| 5 | | FAN COIL UNIT 14 | 1080 | M | 1 | 20 | C | 20 | 1 | M | 1080 | FAN COIL UNIT 9 | | 6 |
| 7 | | SPARE | | | 1 | 20 | A | 15 | 1 | M | 800 | FAN COIL UNIT 4 | | 8 |
| 9 | | FAN COIOL UNITS 10/11 | 1220 | M | 1 | 20 | B | 15 | 1 | M | 800 | FAN COIOL UNIT 5 | | 10 |
| 11 | | FAN COIL UNIT 15 | 1080 | M | 1 | 20 | C | 15 | 1 | M | 800 | FAN COIL UNIT 6 | | 12 |
| 13 | | FAN COIL UNIT 16 | 1080 | M | 1 | 20 | A | 20 | 1 | M | 1080 | FAN COOIL UNIT 1 | | 14 |
| 15 | | SPARE | | | 1 | 20 | B | 20 | 1 | M | 1080 | FAN COIL UNIT 2 | | 16 |
| 17 | | SPARE | | | 1 | 20 | C | 20 | 1 | M | 1080 | FAN COIL UNIT 3 | | 18 |

| LOAD TYPES | CONNECTED LOAD (VA) A | B | C | SUBFEED TOTAL | PANEL TOTAL | DEMAND LOAD (VA) FACTOR | TOTAL |
|---|---|---|---|---|---|---|---|
| LIGHTING | 0 | 0 | 0 | 0 | 0 | 125% | 0 |
| RECEPTACLES | 0 | 0 | 0 | 0 | 0 | - - | 0 |
| MOTORS | 3060 | 4180 | 4040 | 0 | 11280 | 100% | 11280 |
| LARGEST MOTOR | 1080 | 1080 | 1080 | 0 | 3240 | 125% | 4050 |
| EQUIPMENT | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| APPLIANCES | 0 | 0 | 0 | 0 | 0 | 100% | 0 |
| | | | | | **TOTAL DEMAND LOAD (KVA) =** | | **15.3** |
| | | | | | **TOTAL DEMAND LOAD (A) =** | | **43** |

**GENERAL NOTES:**

A. EVERY CIRCUIT AND CIRCUIT MODIFICATION SHALL BE UNIQUELY IDENTIFIED WITH ITS SPECIFIC PURPOSE AND/OR EQUIPMENT AS NECESSARY TO DISTINGUISH FROM ALL OTHER LOADS. IDENTIFICATION SHALL INCLUDE A DEGREE OF DETAIL AS APPROVED BY THE AHJ.

B. PROVIDE NEW UPDATED PANEL DIRECTORIES FOR PANELS THAT ARE INCLUDED IN THIS PROJECT.

C. INDICATE SPARES IN PENCIL.

D. ALL NEW CIRCUIT BREAKERS SHALL MATCH EXISTING PANELBOARD MANUFACTURER AND AIC RATINGS.

STAMP

COLORADO LICENSED TODD SPACEK 45310 PROFESSIONAL ENGINEER
11.03.2026


ST+B ENGINEERING
(303) 900-8782
HTTP://ST8.ENGINEERING
25188 GENESEE TRAIL RD, SUITE 120, GOLDEN, CO 80401


JEFFERSON COUNTY COLORADO
JEFFCO.US (303) 271-6511
100 JEFFERSON CONTY PKWY GOLDEN, CO 80419

# CHILLER REPLACEMENT

JEFFERSON COUNTY - BOETTCHER MANSION
LOOKOUT MOUNTAIN, 900 COLOROW RD, GOLDEN, CO 80401

ST+B PROJECT NO.: 2864.18

| NO | SUBMISSION | DATE |
|---|---|---|
| 1 | 90% CHECK SET | 09/11/26 |
| 1 | 100% CD SET | 09/22/26 |
| 2 | ADDENDUM 1 | 11/03/26 |

SHEET NAME
ELECTRICAL DETAILS

SHEET NO.
E-500


24x36 Z:\Box\ST8 Projects\2864.18 - Jeffco - Boettcher Mansion Equipment Replacement\60D_Drawings\Chiller\E500_Dhcline.dwg
11/3/2026 1:19 PM